Loss Function Summary

Total loss:

$$L = \underbrace{0.5\, L_d}_{\text{depth}} + \underbrace{1.0\, L_n}_{\text{normals}} + \underbrace{0.01\, L_{d\text{-}n}}_{\text{consistency}}$$

$$L_d = L_{\text{PWN}} + L_{\text{VNL}} + L_{\text{silog}} + L_{\text{RPNL}}$$

Pair-wise normal | Virtual normal | Scale-invariant log | Random proposal

$$L_n = 1 - \langle\, \hat{\mathbf{n}},\ \mathbf{n}^* \rangle \qquad \text{(cosine similarity)}$$

$L_{d\text{-}n}$: penalizes disagreement between ∇D (pseudo-normals from depth) and predicted $\mathbf{n}$

Note: $L_{d\text{-}n}$ weight is small (0.01) because pseudo-normals from predicted depth